

Docker vs Virtual Machines

Docker (containers)

- Lightweight => share the host OS kernel
- Start in seconds
- Great for microservices & scaling
- Portable (works the same anywhere)

VMs (virtual machines)

- Heavy => need full OS for each VM
- Start in minutes
- Stronger isolation
- Good for running totally different OSs

Think of it like this: **VMs = full houses**, **Containers = apartments in one building**.

Key Differences

Aspect	Virtual Machine	Docker Container
OS	Full guest OS required	Minimal OS essentials only
Speed	Slow (minutes to hours)	Very fast (1-2 minutes)
Size	Heavy (GBs)	Lightweight (MBs)
Best for	Stable, long-running servers	Quick scaling, rapid deployment

Visual Representation

